THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number that is the square of a prime has exactly: 1

(a) 2 factors **(b)** 3 factors
(c) 4 factors **(d)** an even number of factors

2. A proposition is a statement that is: 1

(a) a question **(b)** either true or false
(c) always false **(d)** always true

3. If a proposition is true, its converse: 1

(a) is never a proposition **(b)** is always true
(c) may be true or false **(d)** is always false

4. 'n is a perfect square' and 'n has an odd number of factors' are related as: 1

(a) only the first implies the second **(b)** only the second implies the first
(c) each implies the other **(d)** neither implies the other

5. **Assertion (A):** If a proposition is true, its converse must also be true. 1
Reason (R): The proposition "If a number is a multiple of 6, then it is a multiple of 3" is true, but 9 is a multiple of 3 and not of 6.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

6. Marks of 82, 74, 90 are combined with weights 1 : 2 : 2. The weighted average is: 1

(a) 83 **(b)** 82 **(c)** 81 **(d)** 84

7. The average of 12 values is 85. When one more value, 33, is added, the new average is: 1

(a) 83 **(b)** 82 **(c)** 80 **(d)** 81

8. Averaging the average heights of 8 seniors and 3 juniors directly does not give the group average because: 1

(a) the seniors are taller **(b)** the two groups have different sizes
(c) averages cannot be combined **(d)** the juniors are shorter

9. A house used 200, 300, 100 and 400 units of electricity on lighting, cooling, kitchen and other appliances in 2010. In a 100% stacked bar for 2010, the length of the cooling segment is: 1

(a) 60% **(b)** 25% **(c)** 30% **(d)** 35%

10. **Assertion (A):** In a 100% stacked bar chart, a longer segment for Bakery A than for Bakery B in the bread category means that A sold more bread than B. 1
Reason (R): A 100% stacked bar shows the percentage share within each bakery's own sales, not the actual amounts.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the converse of "If n is the square of a prime, then n has exactly 3 factors." Give a counterexample if it is false. 2

12. A student says "If a number is divisible by 6, then it is divisible by 2 and 3." Write the converse and state whether both are true. 2

13. What is the difference between a stacked bar chart and a 100% stacked bar chart? 2

14. A bar in a 100% stacked chart shows 25% for lighting. Total consumption is 800 units. How many units are for lighting? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Use the converse of the Baudhāyana–Pythagoras theorem to find which of these triples can be the sides of a right triangle: 3
(i) 8, 15, 17
(ii) 10, 24, 27.

16. A state generated 600, 100, 200 and 100 MW from coal, solar, wind and hydro in 2015. Find the percentage shares for a 100% stacked bar. 3

17. Write the converse of "If a triangle is isosceles, then it has two equal angles". Is it true? Briefly justify using an altitude. 3

OR

Give two examples (one from number theory and one from geometry) of propositions that are true but whose converses are false. 3

18. Explain why a stacked bar chart makes it harder to read the exact value of a middle segment than a cluster-column chart. 3

OR

Bakery A: bread 45%, cakes 35%, snacks 20% of its sales; Bakery B: 30%, 30%, 40%. Which can be inferred: 3
(i) A sold more bread than cakes,
(ii) A sold more bread than B?

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Test whether the triples 5
(a) 15, 20, 25 and
(b) 11, 60, 62 can form right-angled triangles.
(ii) Explain why a triple that satisfies $a^2 + b^2 = c^2$ must form a right-angled triangle.

OR

Show that the following propositions are false by giving a counterexample: 5
(i) If n is an odd number, then $n^2 + 2$ is prime.
(ii) If n is a prime, then $n^2 - 1$ is a multiple of 4.
(iii) If n is divisible by 9, then it is divisible by 27.

20. (i) A shop mixes 5 kg of tea at ₹200 per kg with 3 kg of tea at ₹280 per kg. Find the price per kg of the blend. 5
(ii) At what price should the blend be sold to earn 20% profit?

OR

(i) State the difference in what can be inferred from stacked and 100% stacked bar charts. 5
(ii) Give one question easier to answer with each.

END OF QUESTION PAPER — Total: 40 Marks